HW Assignment (Midterm Substitute)

Derivative Securities (주식, 채권, 파생금융상품 3) · Prof. Dong-Hyun Ahn · Fall 2026

경제학부 2022-15364 김은수 November 11, 2026

Question 1 — Margin call and withdrawal

> 2 long futures contracts on frozen orange juice, 15,000 lb each, futures price 160 ¢/lb, initial margin $6,000/contract, maintenance margin $4,500/contract. What price change leads to a margin call? When could $2,000 be withdrawn?

The position covers $2 \times 15{,}000 = 30{,}000$ lb, so a change of 1 ¢/lb in the futures price changes the margin balance by $30{,}000 \times \$0.01 = \$300$.

- Initial margin for the position: $2 \times \$6{,}000 = \$12{,}000$
- Maintenance margin for the position: $2 \times \$4{,}500 = \$9{,}000$

(a) Margin call

A margin call occurs when the balance falls below the maintenance margin, i.e. when the cumulative loss on the long position exceeds $\$12{,}000 - \$9{,}000 = \$3{,}000$:

$$30{,}000 \times \Delta F < -\$3{,}000 \implies \Delta F < -\$0.10 \text{ per lb}.$$

Answer. A margin call is triggered if the futures price **falls by more than 10 ¢/lb**, i.e. below **150 ¢/lb**. The trader must then deposit variation margin to restore the balance to the initial margin of $12,000.

(b) Withdrawal of $2,000

Only the balance in excess of the initial margin can be withdrawn. To withdraw $2,000 the balance must reach at least $\$12{,}000 + \$2{,}000 = \$14{,}000$, so the cumulative gain must be at least $2,000:

$$30{,}000 \times \Delta F \geq \$2{,}000 \implies \Delta F \geq \$0.0667 \text{ per lb}.$$

Answer. $2,000 can be withdrawn if the futures price **rises by at least 6.67 ¢/lb**, i.e. to **166.67 ¢/lb** or higher (assuming no margin call has occurred in between).

Question 2 — Minimum-variance hedge ratio

> $\sigma_S = 0.65$, $\sigma_F = 0.81$ (quarterly changes), $\rho = 0.8$. What is the optimal hedge ratio for a three-month contract? What does it mean?

The variance of the hedged position, $v = \sigma_S^2 + h^2\sigma_F^2 - 2h\rho\sigma_S\sigma_F$, is minimised where $dv/dh = 0$ (with $d^2v/dh^2 = 2\sigma_F^2 > 0$):

$$h^* = \rho\,\frac{\sigma_S}{\sigma_F} = 0.8 \times \frac{0.65}{0.81} = \mathbf{0.642}.$$

The hedge lasts three months and the volatilities are for quarterly changes, so the data match the life of the hedge.

What it means. To minimise the variance of the hedged position, the hedger should take a futures position equal to **64.2% of the size of the exposure** (e.g. futures on 642 units for every 1,000 units of the commodity), not 100%. Equivalently, h^* is the slope β_S from regressing ΔS on ΔF: a $1 change in the futures price comes, on average, with a $0.642 change in the spot price. The futures price is more volatile than the spot price ($0.81 > 0.65$) and the two are not perfectly correlated, so a 1:1 hedge would over-hedge.

Hedge ratio	Variance of $\Delta S - h\Delta F$	Std. dev.
No hedge, $h = 0$	$0.65^2 = 0.4225$	\$0.650
Naïve hedge, $h = 1$	$0.4225 + 0.6561 - 2(0.8)(0.65)(0.81) = 0.2362$	\$0.486
Optimal hedge, $h^* = 0.642$	$\sigma_S^2(1-\rho^2) = 0.4225 \times 0.36 = 0.1521$	\$0.390

The hedge effectiveness is $\rho^2 = 0.64$: the optimal hedge removes 64% of the variance of the unhedged exposure, and it gives a lower variance than the naïve 1:1 hedge.

Question 3 — "Speculation in futures markets is pure gambling"

> "Speculation in futures markets is pure gambling. It is not in the public interest to allow speculators to buy seats on a futures exchange." Discuss the viewpoint.

The viewpoint is too strong. A speculator, like a gambler, takes a risky position in the hope of a profit, but futures speculation does something useful for the economy that gambling does not do.

1. **Speculators are the counterparties hedgers need.** Hedgers rarely balance out: there may be many more producers wanting to sell futures than processors wanting to buy them. Speculators take the other side and so **carry the price risk that hedgers want to get rid of**. Gambling creates risk that did not exist before (the outcome of a game); futures speculation takes on risk that already exists in the economy (the future price of oil or wheat) and moves it to those most willing to bear it. Speculators are paid for this through an expected risk premium (Keynes and Hicks' *normal backwardation*, $F_0 < E[S_T]$).
2. **Liquidity and lower hedging costs.** Active speculation means more trading volume and open interest and narrower bid–ask spreads. Hedgers can then open and close large positions quickly and cheaply, and this matters a lot when they **roll hedges forward** (Question 7).
3. **Price discovery.** Speculators gather and trade on information about future supply and demand. This makes futures prices more informative, and producers, consumers and policymakers use them as forecasts when deciding on production, storage and investment.

Valid concerns. Speculation can be harmful when it turns into **manipulation** (e.g. attempts to corner a market, such as the Hunt brothers in silver, 1979–80), when it uses too much **leverage**, or when it adds to price bubbles and volatility. Barings, Metallgesellschaft and LTCM, discussed in class, show how leveraged positions and margin calls can cause very large losses and spill over to other institutions.

Conclusion. These problems call for **regulation, not a ban**: position limits, margin requirements and daily marking to market, the clearing house, reporting rules, and supervision by regulators (the CFTC in the U.S.). Keeping speculators out of the exchange would make futures markets thinner and less informative, and hedging would become more expensive, which would hurt the hedgers the restriction is meant to protect. So allowing speculators to trade, under suitable rules, is in the public interest.

Question 4 — Choice of contract month

> CBOT corn futures delivery months: March, May, July, September, December. Which contract should be used when the hedge expires in (a) June, (b) July, (c) January?

Rule. Choose the contract with the **earliest delivery month that is later than the month in which the hedge expires**, and do not use a contract that matures in the same month as the hedge. Two reasons:

- Futures prices are often erratic during the delivery month, and a long hedger holding the contract in the delivery month may have to take physical delivery (the short party decides when to deliver).
- A contract that matures soon after the hedge ends keeps the basis risk small, because the spot price and the futures price converge as maturity approaches.

Hedge expires in	Contract to use	Reason
(a) June	**July**	First delivery month after June
(b) July	**September**	The July contract matures in the hedge month itself, so use the next one
(c) January	**March**	The December contract has already expired; March (next year) is the next one

Question 5 — Basis risk for a short hedger

> With $b = S - F$, the short hedger's effective selling price is $S_T + (F_0 - F_T) = F_0 + b_T$. Explain why the position improves when the basis strengthens unexpectedly and worsens when it weakens unexpectedly.

The short hedger is **long the asset and short futures**. The value of the combined position at T is

$$\underbrace{S_T}_{\text{sale of asset}} + \underbrace{(F_0 - F_T)}_{\text{gain on short futures}} = F_0 + (S_T - F_T) = F_0 + b_T.$$

F_0 is known at the start of the hedge. The only uncertain part is the basis at the close-out date, b_T, so the short hedger is effectively **long the basis**. When the hedge is set up, the hedger expects to receive $F_0 + E[b_T]$.

- **Basis strengthens unexpectedly** ($b_T > E[b_T]$): the spot price rises by more, or falls by less, than the futures price. The gain on the asset is more than the loss on the short futures (or the loss on the asset is less than the gain on the futures). The effective price $F_0 + b_T$ is then above what was expected, so the hedger is **better off**.
- **Basis weakens unexpectedly** ($b_T < E[b_T]$): the spot price rises by less, or falls by more, than the futures price. The futures leg no longer fully offsets the asset leg, and the effective price is below what was expected, so the hedger is **worse off**.

Example. $F_0 = \$50$ and the hedger expects $b_T = -\$1$, so the expected selling price is $\$49$.

Outcome	S_T	F_T	b_T	Futures gain $F_0 - F_T$	Effective price
As expected	45	46	−1	+4	49
Basis strengthens	47	46	+1	+4	**51** (better)
Basis weakens	43	46	−3	+4	**47** (worse)

For a **long hedger** (short the asset, long futures) the effective purchase price is also $F_0 + b_T$, but it is a cost, so a long hedger gains when the basis weakens and loses when it strengthens. (Note: the lecture notes also define the basis as $F - S$. With that definition every sign above is reversed, but the conclusion does not change.)

Question 6 — Hedging a cattle purchase

> Monthly $\sigma_S = 1.2$ ¢/lb, $\sigma_F = 1.4$ ¢/lb, $\rho = 0.7$. On Oct 15 a beef producer commits to buy 200,000 lb of live cattle on Nov 15 and hedges with December live cattle futures (40,000 lb per contract). What strategy should the producer follow?

The producer will **buy** cattle in one month, so it loses if prices rise and needs a **long hedge**. The hedge lasts one month (Oct 15 → Nov 15), which matches the monthly volatilities given. The minimum-variance hedge ratio is

$$h^* = \rho \frac{\sigma_S}{\sigma_F} = 0.7 \times \frac{1.2}{1.4} = 0.6.$$

Optimal number of contracts:

$$N^* = \frac{h^* \times Q_A}{Q_F} = \frac{0.6 \times 200{,}000}{40{,}000} = 3.$$

Answer. On October 15 the producer should **go long 3 December live cattle futures contracts** (120,000 lb). On November 15 it buys the 200,000 lb of cattle in the spot market and **closes out** the 3 contracts by selling them. The December contract is the nearest one that is still trading after November 15.

Result: if cattle prices rise, the higher purchase cost is partly offset by the gain on the long futures. The standard deviation of the one-month change in the cost falls from $1.2 \times 200{,}000 = 240{,}000¢ = \$2{,}400$ (unhedged) to $\sqrt{1.2^2(1-0.7^2)} \times 200{,}000 \approx 0.857 \times 200{,}000¢ \approx \$1{,}714$. Some **basis risk** remains, because the December futures price will not move exactly with the November 15 spot price.

Question 7 — Rolling the hedge forward

> 100,000 bbl to be sold in June 2027, hedge ratio 1.0, spot \$19 in April 2026. Short 100 Oct-2026 contracts, rolled into Mar-2027 (Sept 2026) and into Jul-2027 (Feb 2027). Spot falls from \$19 to \$16. Gains from the roll? How much protection against the \$3 decline? Effect of a hedge ratio of 1.5?

(a) Total gains from rolling the hedge

Contract	Shorted at	Closed out at	Gain per bbl	Gain (100,000 bbl)
Oct 2026 (Apr 2026 → Sep 2026)	\$18.20	\$17.40	\$0.80	\$80,000
Mar 2027 (Sep 2026 → Feb 2027)	\$17.00	\$16.50	\$0.50	\$50,000
Jul 2027 (Feb 2027 → Jun 2027)	\$16.30	\$15.90	\$0.40	\$40,000
Total			**\$1.70**	**\$170,000**

Answer. Total gain on the futures = **\$1.70 per barrel = \$170,000**.

(b) Protection against the \$3 decline

The oil loses $\$19 - \$16 = \$3$ per barrel, i.e. \$300,000 on 100,000 barrels. The futures gain of \$1.70/bbl offsets $1.70/3.00 \approx \mathbf{56.7\%}$ of this loss. The effective selling price is

$$S_T + \text{futures gains} = \$16.00 + \$1.70 = \mathbf{\$17.70} \text{ per barrel},$$

i.e. \$1.77 million in total, compared with \$1.60 million without the hedge.

Why not 100%? (Hint: basis risk.) A futures hedge locks in the **initial futures price** $F_0 = \$18.20$, not the April spot price of \$19. The \$0.80 gap between them is the initial basis: the oil curve is in backwardation, and this gap is already known in April 2026. The effective price can be split into the initial futures price, the rollover basis at each roll and the final basis:

$$\underbrace{18.20}_{F_0} + \underbrace{(17.00 - 17.40)}_{\text{1st rollover basis}=-0.40} + \underbrace{(16.30 - 16.50)}_{\text{2nd rollover basis}=-0.20} + \underbrace{(16.00 - 15.90)}_{\text{final basis } S_T - F_T = +0.10} = 17.70.$$

At each roll the new (more distant) contract was **cheaper** than the contract being closed out. A short hedger rolling forward therefore lost \$0.60/bbl in **rollover basis**, and gained \$0.10/bbl back on the final (classic) basis.

(c) Hedge ratio of 1.5

The company now shorts $1.5 \times 100 = 150$ contracts at each stage (150,000 bbl):

$$\text{Futures gain} = 1.5 \times \$1.70 = \$2.55/\text{bbl}\ (\$255{,}000), \qquad \text{Effective price} = \$16.00 + \$2.55 = \mathbf{\$18.55}/\text{bbl}.$$

In this outcome the larger hedge offsets $2.55/3.00 = 85\%$ of the \$3 decline and gives \$85,000 more than the 1.0 hedge. But this is only because prices fell. With $h = 1.5$ the company is **over-hedged**: on top of a full hedge it holds a net short futures position on 50,000 barrels, which is a speculative bet that prices will fall. Its value is now

$$S_T + 1.5 \sum (F_{\text{open}} - F_{\text{close}}),$$

which *falls* when oil prices rise. If prices had risen instead, the loss on the 150 short contracts would have been larger than the gain on the oil, and the company would have been worse off than with $h = 1.0$. The rollover-basis losses ($0.60 \rightarrow 0.90$/bbl) also grow with the size of the futures position. A hedge ratio other than 1.0 is justified only if it is close to the minimum-variance ratio $h^* = \rho \sigma_S / \sigma_F$ (Question 2), and that requires σ_F to be much smaller than σ_S. For oil hedged with oil futures, where spot and futures move closely together, h^* is close to 1, so a ratio of 1.5 adds risk instead of reducing it.

Question 8 — Currency futures arbitrage (Swiss franc)

> 2-month rates (continuous): $r_{CHF} = 3\%$, $r_{USD} = 8\%$. Spot $S_0 = \$0.6500$/CHF, 2-month futures $F_0 = \$0.6600$. What arbitrage opportunities does this create?

By covered interest rate parity, the theoretical futures price is

$$F_0^* = S_0\, e^{(r - r_f)T} = 0.65\, e^{(0.08-0.03)\times 2/12} = 0.65 \times 1.008368 = \$0.6554.$$

The market price $\$0.6600 > \$0.6554$, so the futures contract is **overpriced**. Strategy: **sell the futures, buy francs spot with borrowed dollars, and invest the francs at the Swiss rate**. Per 1 CHF delivered at $T = 2/12$:

Transaction	$t = 0$	$t = T$ **(2 months)**
Borrow USD at 8%: $0.65\, e^{-0.03/6} = \$0.64676$	+\$0.64676	$-0.64676\, e^{0.08/6} = -\$0.65544$
Buy $e^{-0.03/6} = 0.99501$ CHF spot and lend it at 3%	−\$0.64676	receive 1 CHF $= S_T$
Short 1 CHF futures at \$0.6600	0	$0.6600 - S_T$
Total	**0**	$0.6600 - 0.65544 = +\$0.00456$

Answer. Arbitrageurs **short Swiss franc futures, borrow dollars at 8%, buy francs spot and invest them at 3%**. With no net investment and no risk, this earns **\$0.00456 per franc** at maturity (≈ \$570 on a standard 125,000-CHF contract). This trading pushes the futures price down and/or the spot price up until $F_0 = \$0.6554$.

Question 9 — Stock index futures arbitrage

> $r = 10\%$ (continuous), dividend yield $q = 4\%$, index $S_0 = 400$, 4-month futures price $F_0 = 405$. What arbitrage opportunities does this create?

The theoretical futures price is

$$F_0^* = S_0\, e^{(r-q)T} = 400\, e^{(0.10-0.04)\times 4/12} = 400\, e^{0.02} = 408.08.$$

The market price $405 < 408.08$, so the futures contract is **underpriced**. Strategy: **buy the futures, short-sell the stocks in the index, and invest the proceeds at the risk-free rate** until delivery. Per unit of the index:

Transaction	$t = 0$	$t = T$ **(4 months)**
Short-sell $e^{-qT} = e^{-0.04/3} = 0.98676$ units of the index (the dividends owed on the short are covered by increasing the short, so it grows to exactly 1 unit at T)	+394.70	buy back 1 unit: $-S_T$
Lend the proceeds at 10%	−394.70	$394.70\, e^{0.10/3} = +408.08$
Long 1 futures at 405	0	$S_T - 405$
Total	**0**	$408.08 - 405 = +3.08$

Answer. Arbitrageurs should **buy index futures and short the stocks underlying the index**, investing the proceeds at 10%. This locks in a riskless profit of about **3.08 index points per unit** at delivery (multiply by the contract multiplier for the dollar amount). In practice, an investor who already holds the index portfolio can simply sell it instead of short-selling. This index arbitrage raises the futures price and/or lowers the index until $F_0 = 408.08$, apart from transaction costs and short-sale constraints.